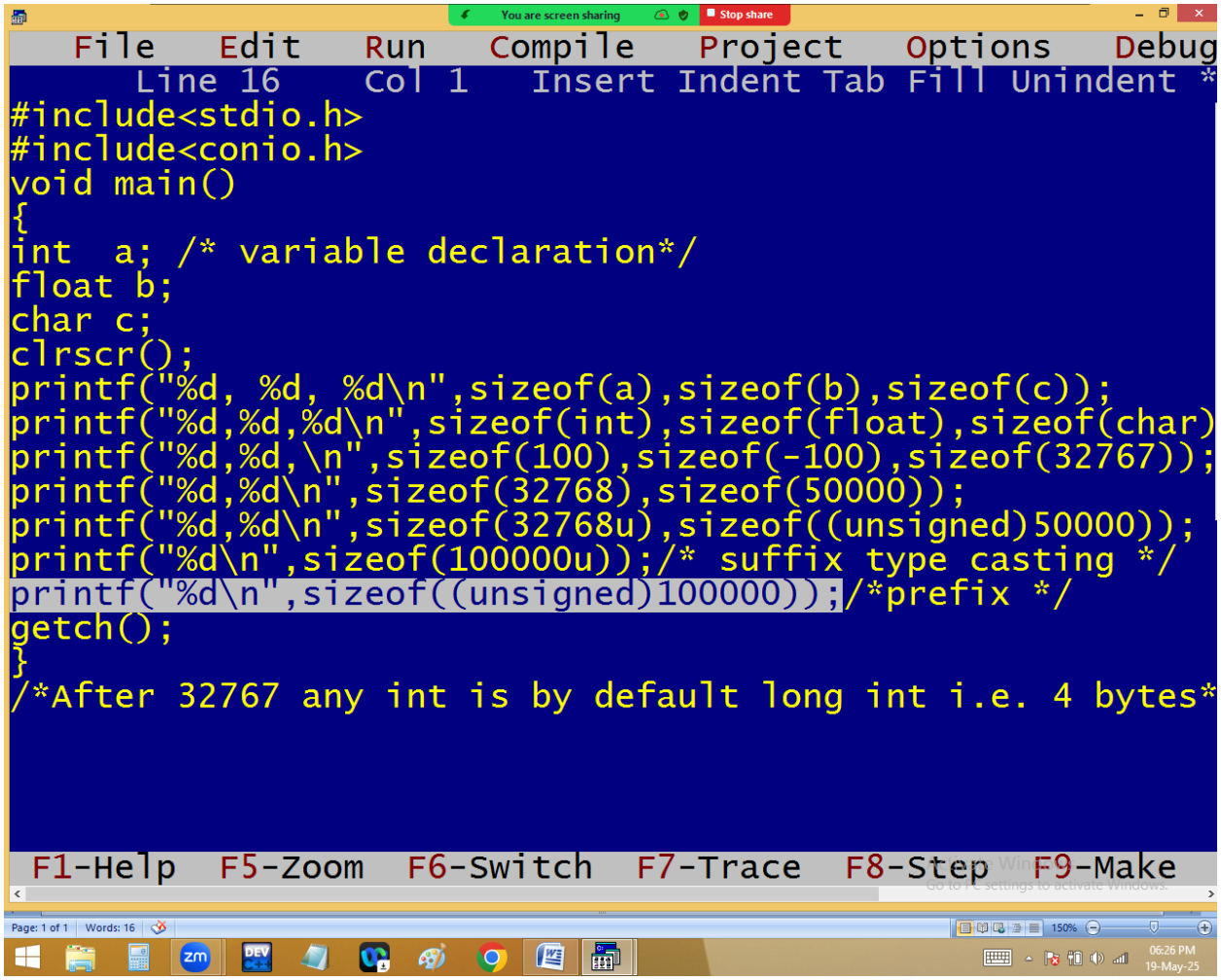
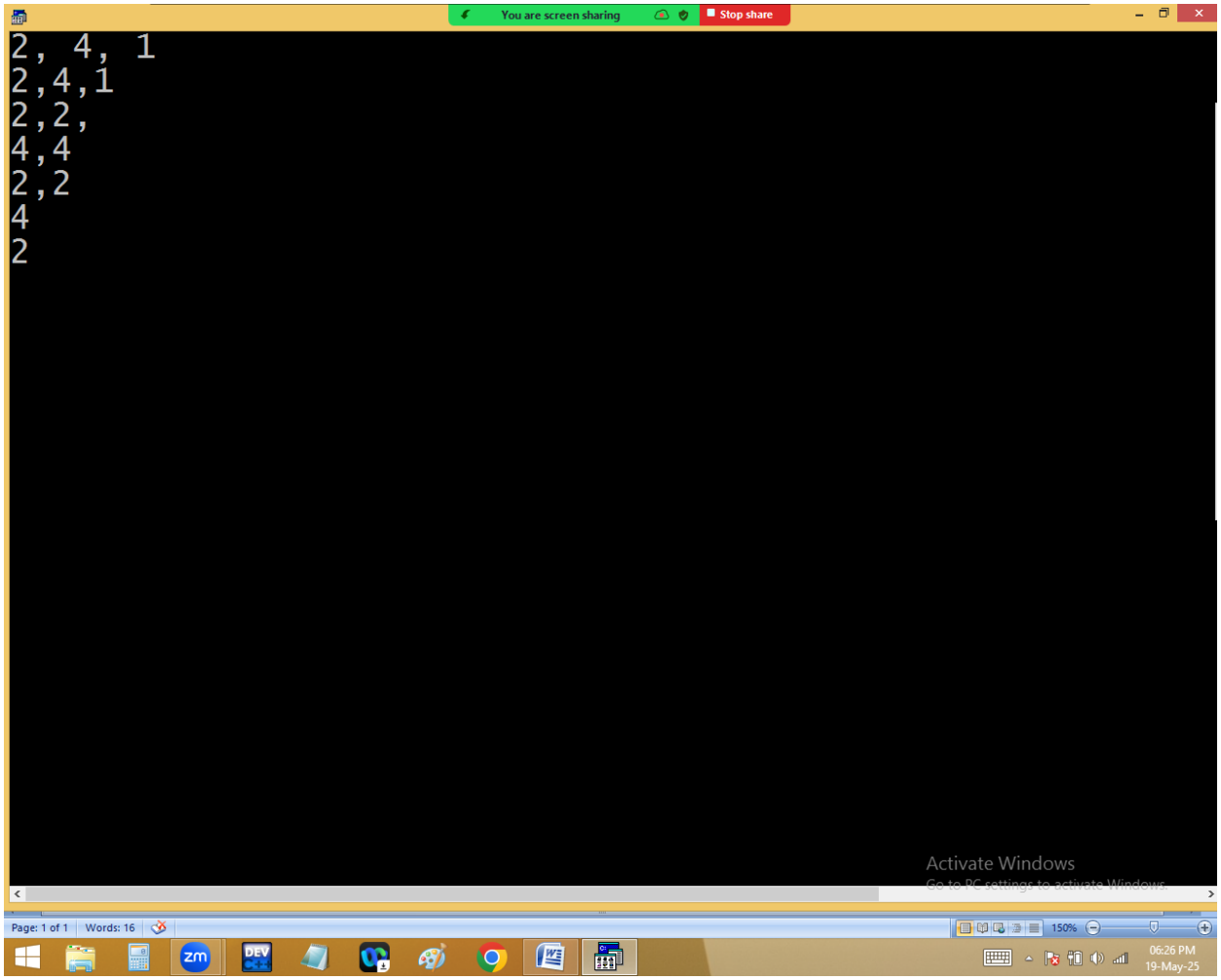
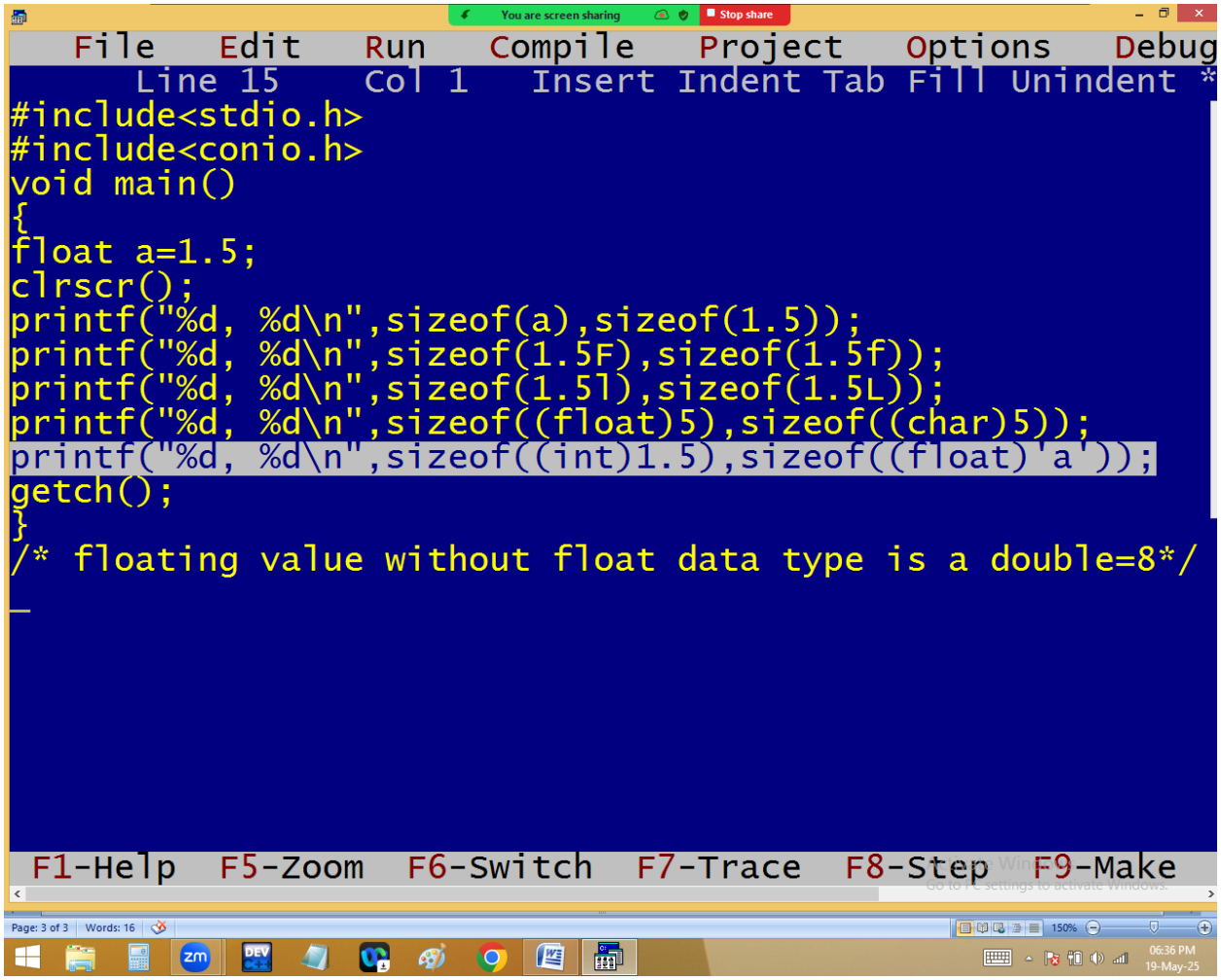
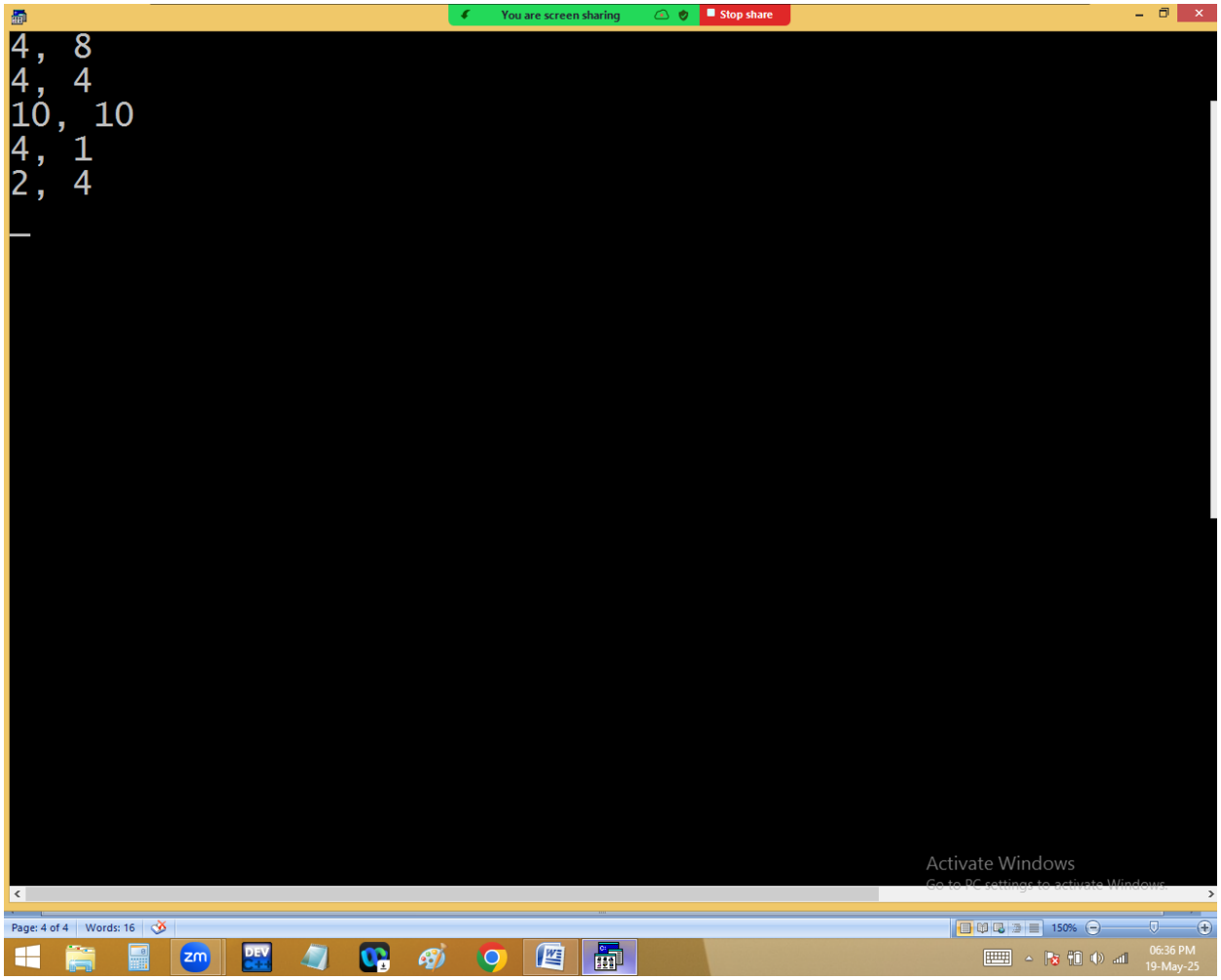
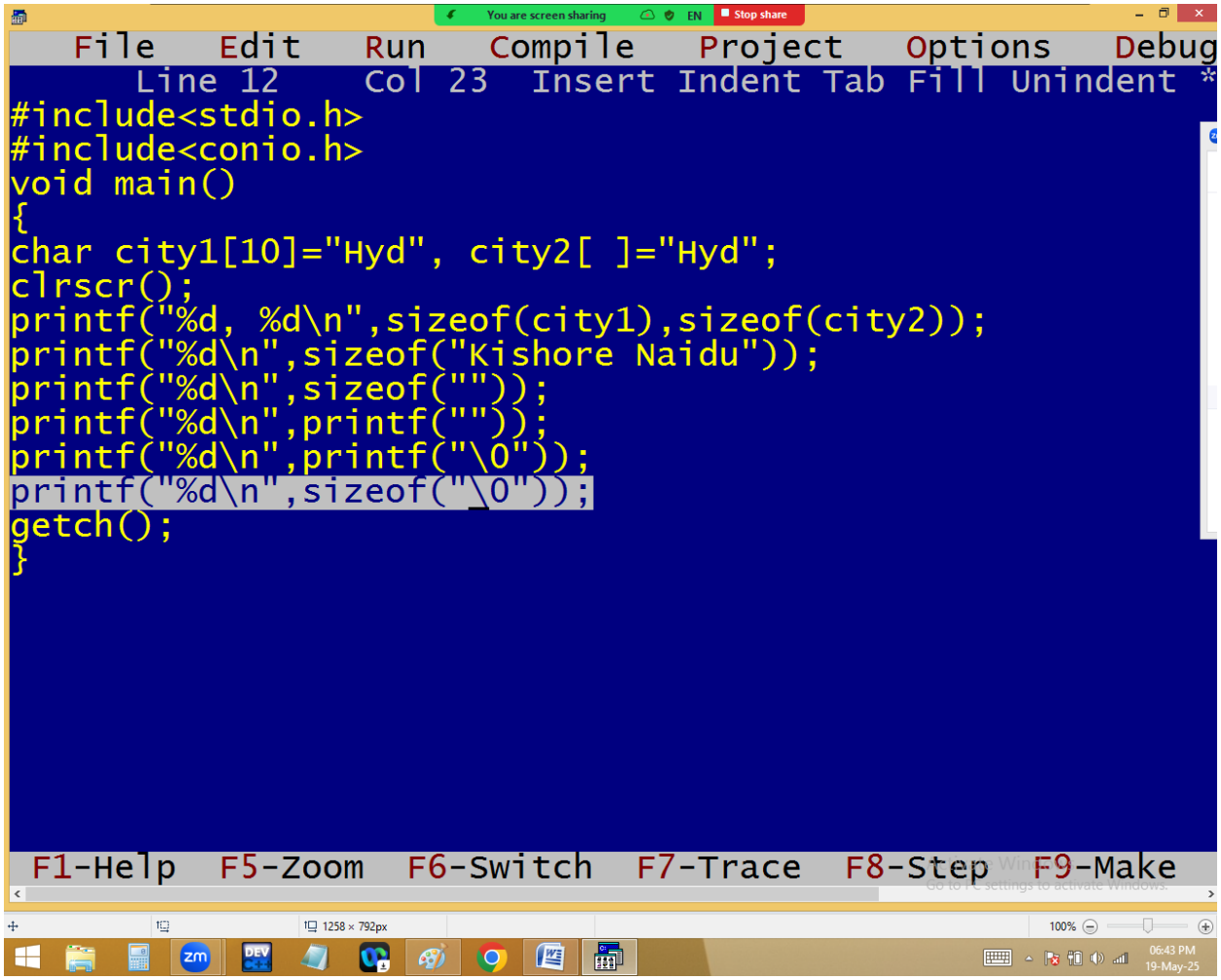
sizeof(): It return the no of bytes required for a variable / data type / value.

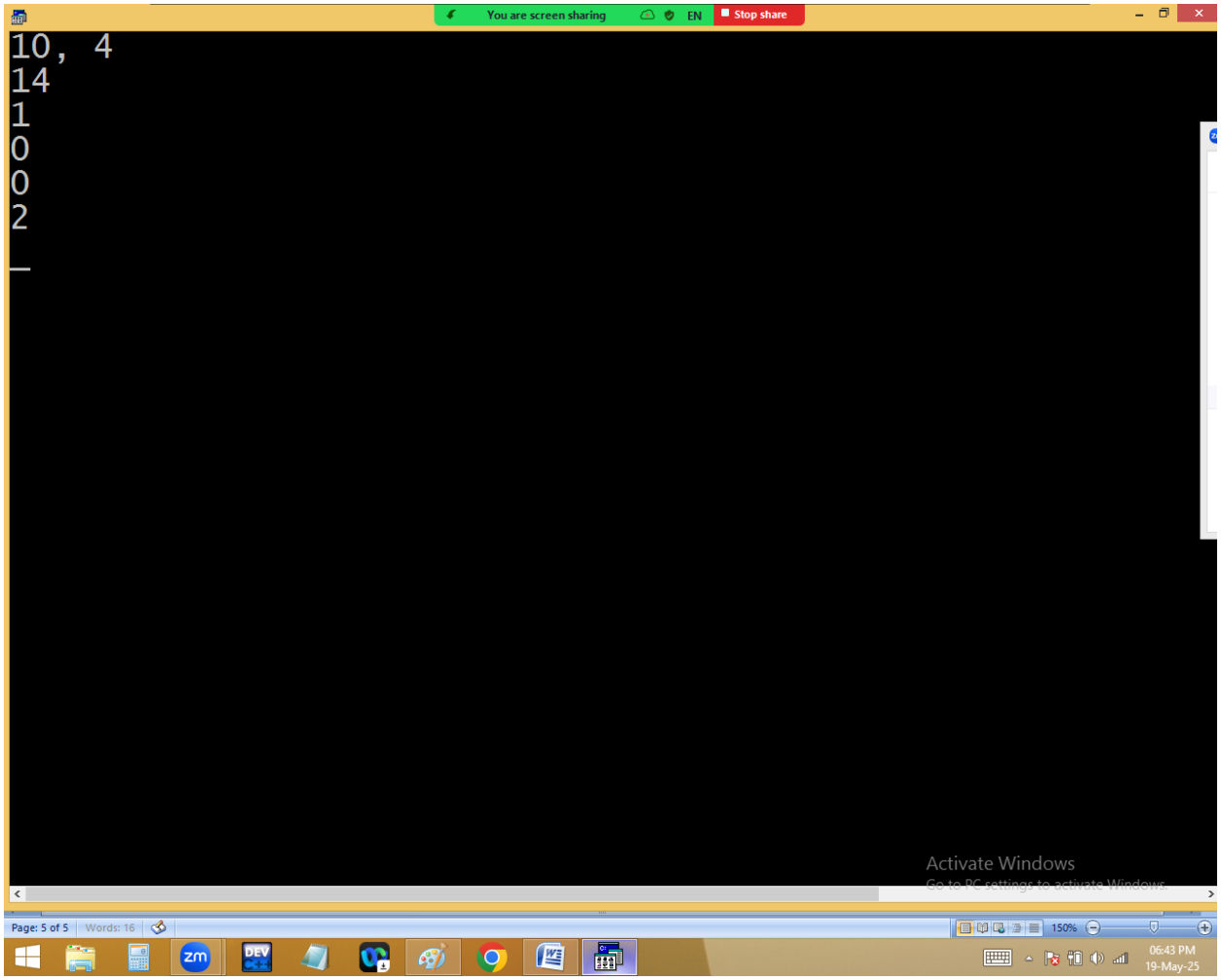

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug
Line 16 Col 1 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* variable declaration*/
float b;
char c;
clrscr();
printf("%d, %d, %d\n",sizeof(a),sizeof(b),sizeof(c));
printf("%d,%d,%d\n",sizeof(int),sizeof(float),sizeof(char)
printf("%d,%d,\n",sizeof(100),sizeof(-100),sizeof(32767));
printf("%d,%d\n",sizeof(32768),sizeof(50000));
printf("%d,%d\n",sizeof(32768u),sizeof((unsigned)50000));
printf("%d\n",sizeof(100000u));/* suffix type casting */
printf("%d\n",sizeof((unsigned)100000));/*prefix */
getch();
}
/*After 32767 any int is by default long int i.e. 4 bytes*
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 1 of 1 Words: 16
150%
06:26 PM
19-May-25

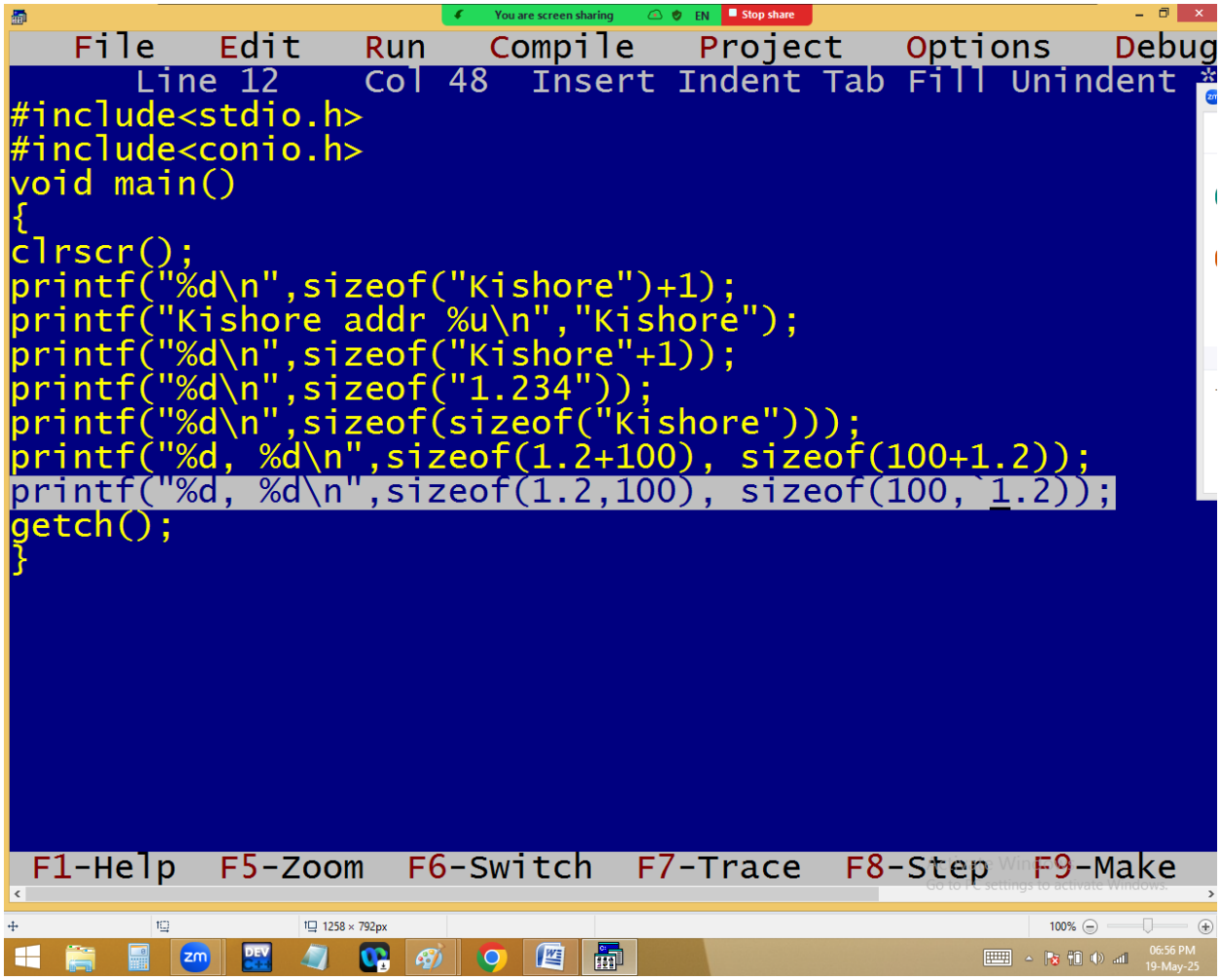

You are screen sharing
Stop share
2, 4, 1
2,4,1
2,2,
4,4
2,2
4
2
Activate Windows
Page: 1 of 1
Words: 16
150%
06:26 PM
19-May-25

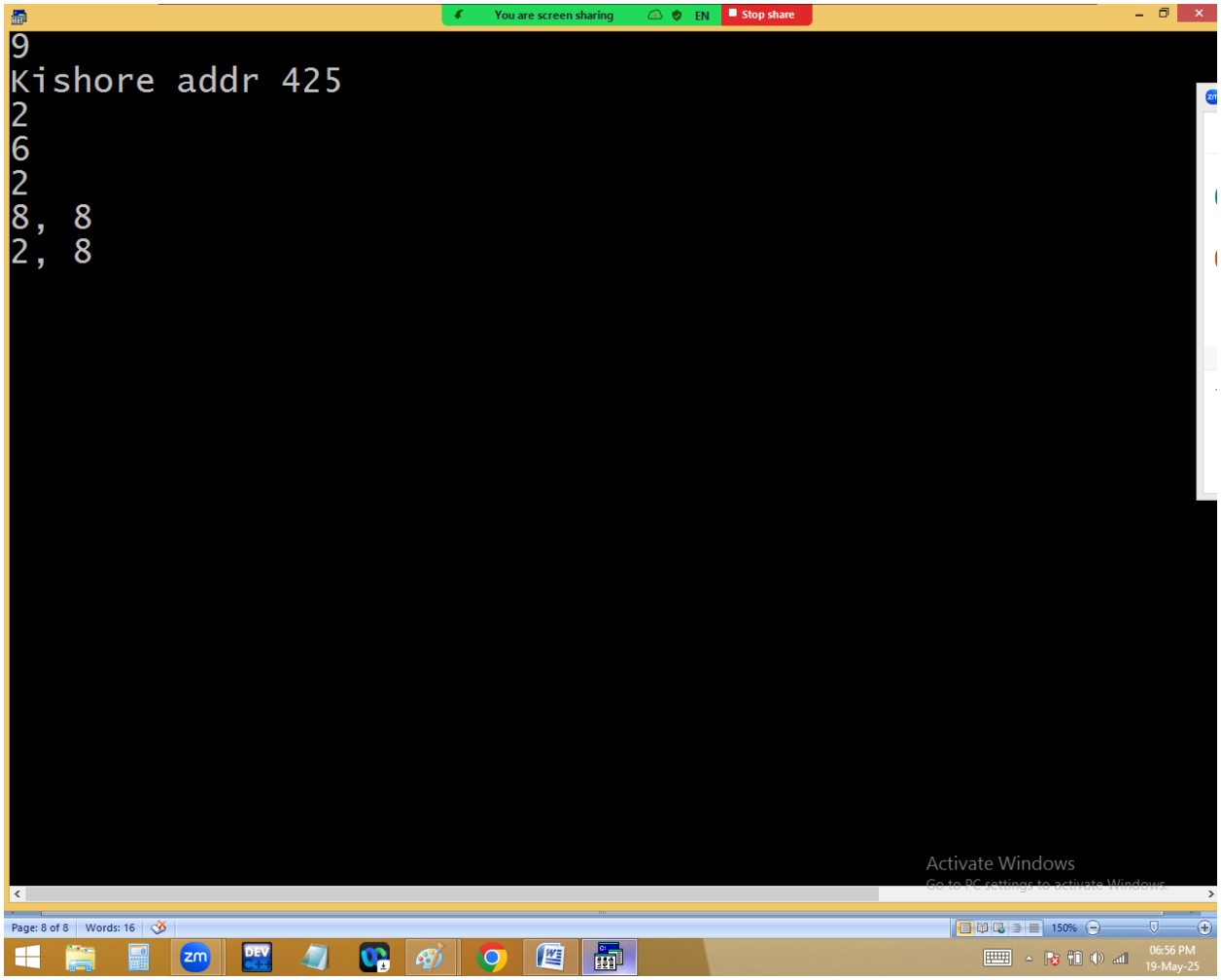

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug
Line 15 Col 1 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
float a=1.5;
clrscr();
printf("%d, %d\n",sizeof(a),sizeof(1.5));
printf("%d, %d\n",sizeof(1.5F),sizeof(1.5f));
printf("%d, %d\n",sizeof(1.5l),sizeof(1.5L));
printf("%d, %d\n",sizeof((float)5),sizeof((char)5));
printf("%d, %d\n",sizeof((int)1.5),sizeof((float)'a'));
getch();
}
/* floating value without float data type is a double=8*/
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 3 of 3
Words: 16
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19-May-25

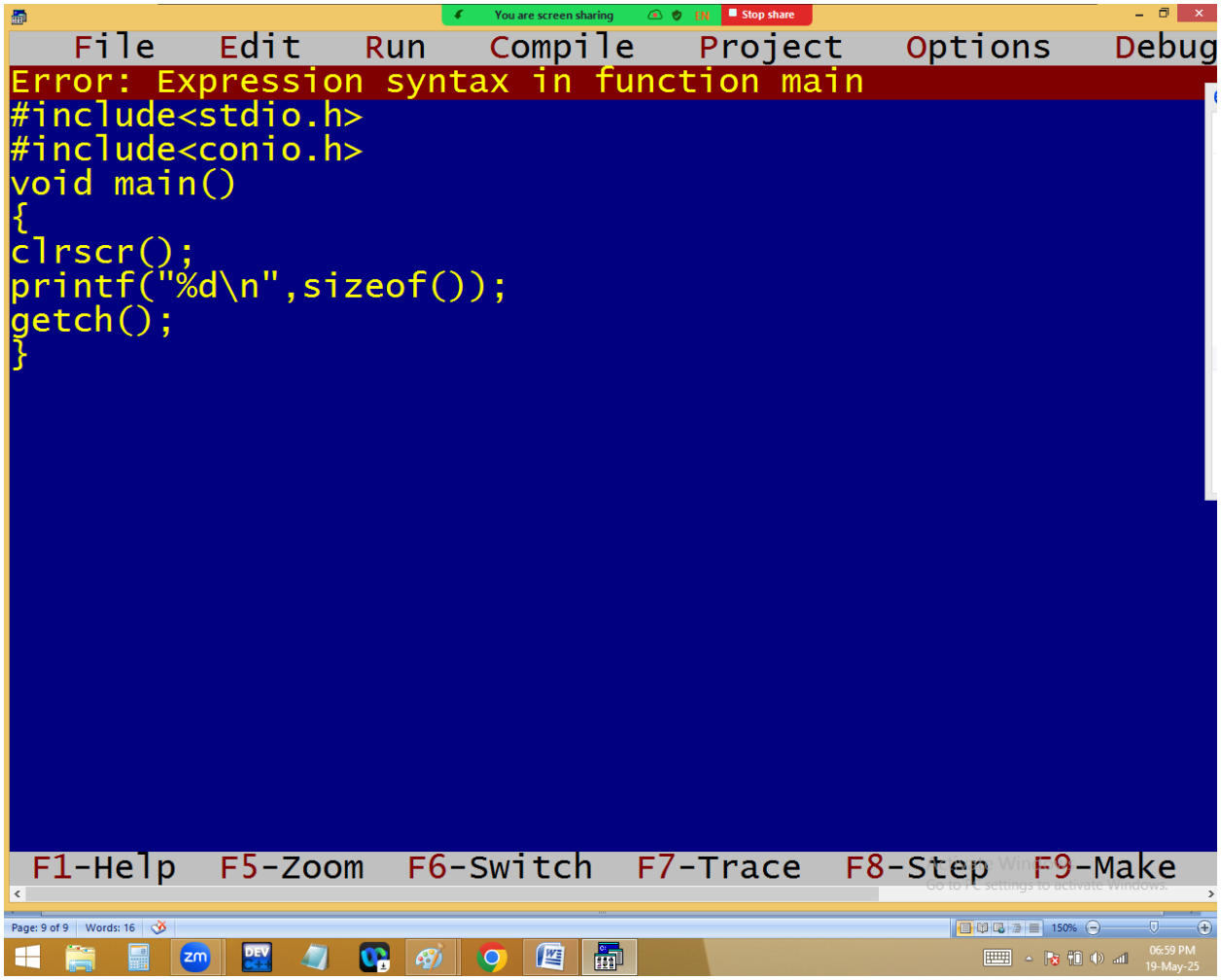

You are screen sharing
Stop share
4, 8
4, 4
10, 10
4, 1
2, 4
Activate Windows
Page: 4 of 4
Words: 16
150%
06:36 PM
19-May-25

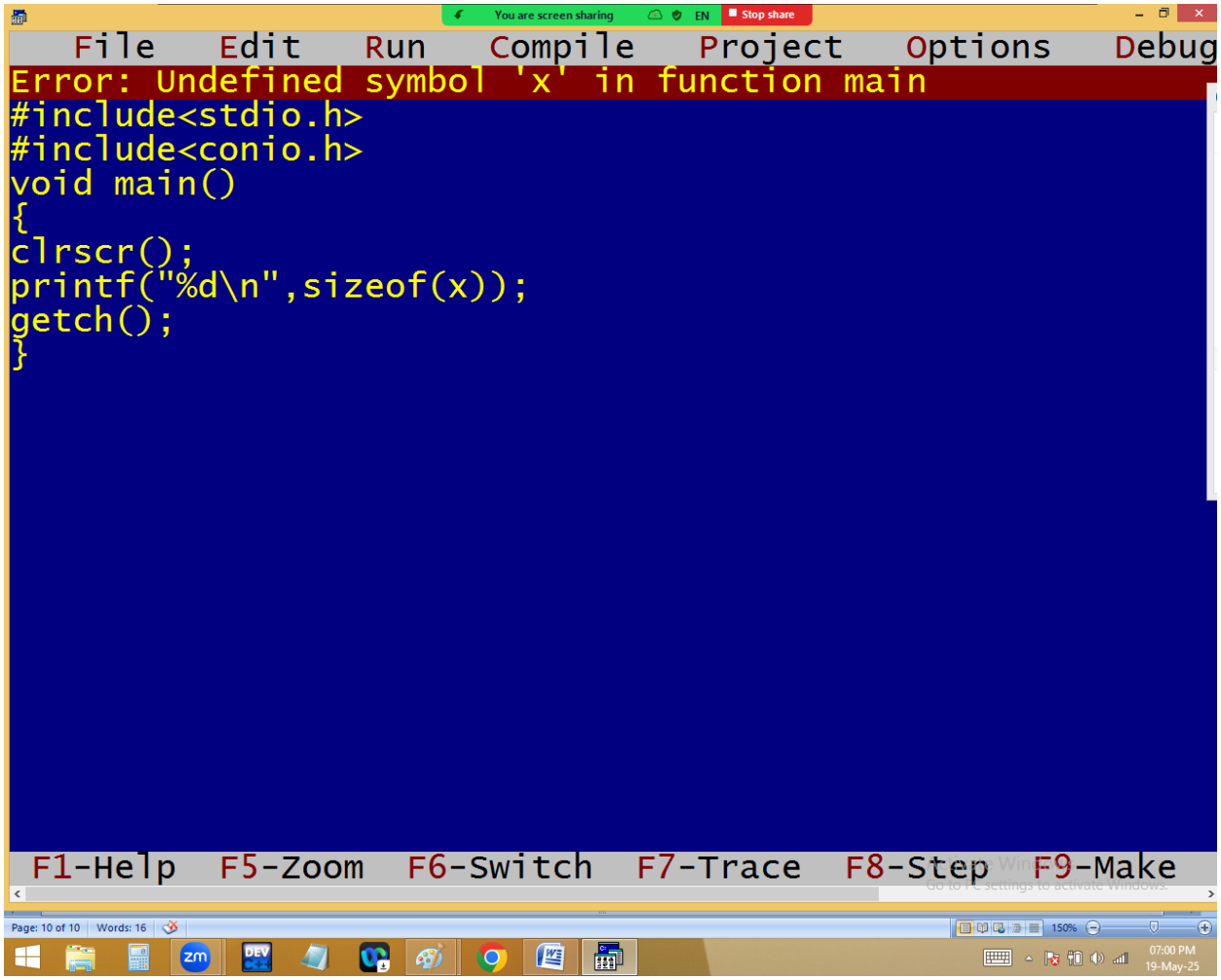

You are screen sharing
EN
Stop share
File Edit Run Compile Project Options Debug
Line 12 Col 23 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
char city1[10]="Hyd", city2[ ]="Hyd";
clrscr();
printf("%d, %d\n",sizeof(city1),sizeof(city2));
printf("%d\n",sizeof("Kishore Naidu"));
printf("%d\n",sizeof(""));
printf("%d\n",printf(""));
printf("%d\n",printf("\0"));
printf("%d\n",sizeof("\0"));
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
1258 × 792px
100%
06:43 PM
19-May-25

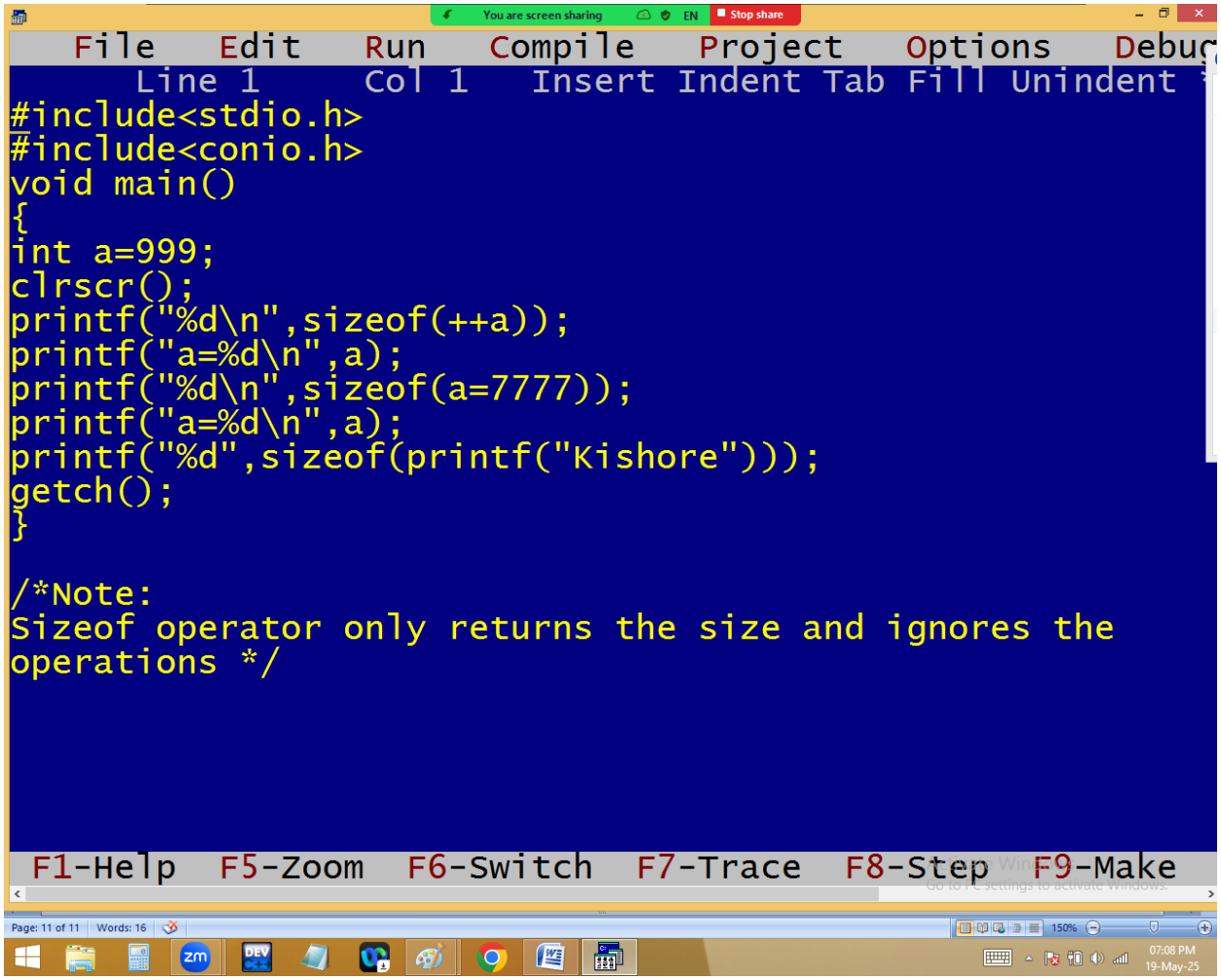

You are screen sharing
EN
Stop share
10, 4
14
1
0
0
2
Activate Windows
Page: 5 of 5
Words: 16
150%
06:43 PM
19-May-25

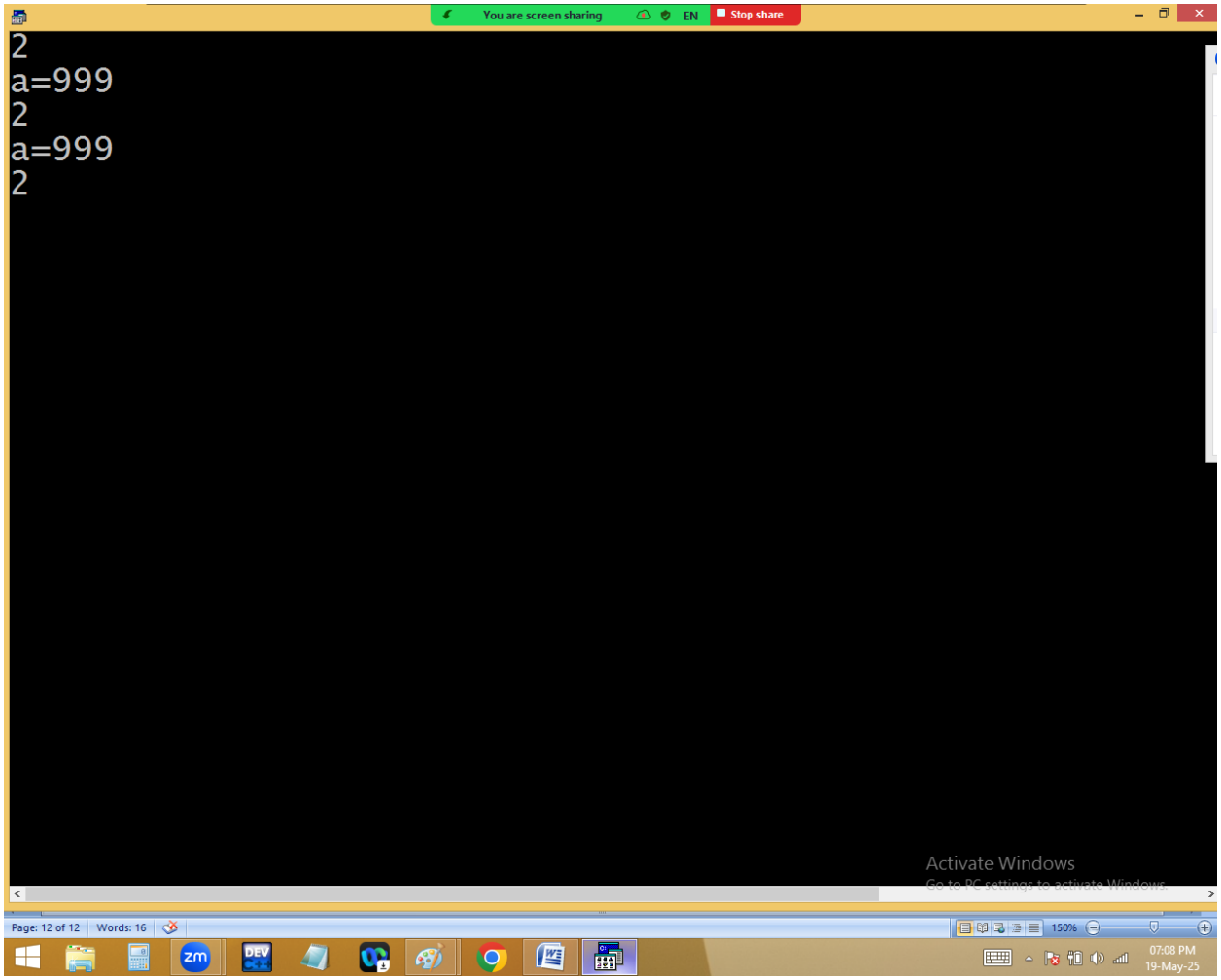

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Stop share
File Edit Run Compile Project Options Debug
Line 12 Col 48 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof("Kishore")+1);
printf("Kishore addr %u\n","Kishore");
printf("%d\n",sizeof("Kishore"+1));
printf("%d\n",sizeof("1.234"));
printf("%d\n",sizeof(sizeof("Kishore")));
printf("%d, %d\n",sizeof(1.2+100), sizeof(100+1.2));
printf("%d, %d\n",sizeof(1.2,100), sizeof(100, 1.2));
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
1258 × 792px
100%
06:56 PM
19-May-25


You are screen sharing
EN
Stop share
9
Kishore addr 425
2
6
2
8, 8
2, 8
Activate Windows
Page: 8 of 8
Words: 16
150%
06:56 PM
19-May-25

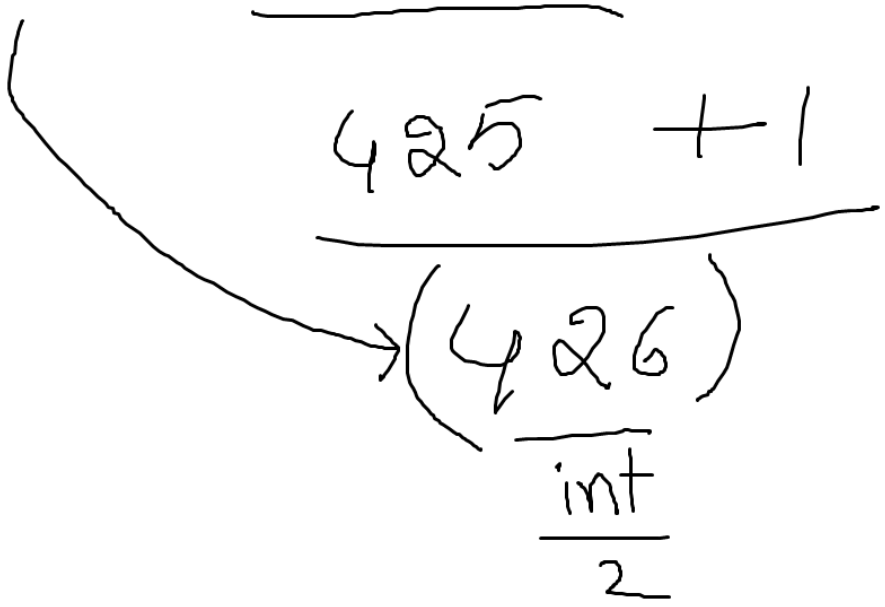

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug
Error: Expression syntax in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof());
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 9 of 9
Words: 16
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19-May-25


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EN
Stop share
File Edit Run Compile Project Options Debug
Error: Undefined symbol 'x' in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof(x));
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 10 of 10
Words: 16
150%
07:00 PM
19-May-25

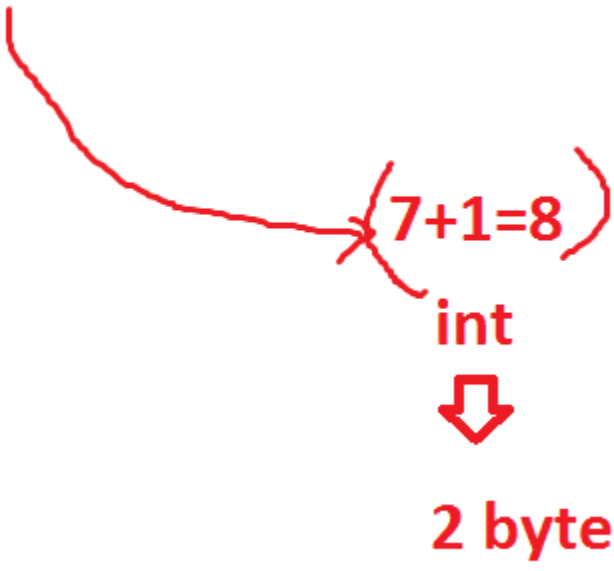

You are screen sharing
EN
Stop share
File Edit Run Compile Project Options Debug
Line 1 Col 1 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a=999;
clrscr();
printf("%d\n",sizeof(++a));
printf("a=%d\n",a);
printf("%d\n",sizeof(a=7777));
printf("a=%d\n",a);
printf("%d",sizeof(printf("Kishore")));
getch();
}
/*Note:
Sizeof operator only returns the size and ignores the
operations */
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make
Page: 11 of 11
Words: 16
150%
07:08 PM
19-May-25


You are screen sharing
EN
Stop share
2
a=999
2
a=999
2
Activate Windows
Page: 12 of 12
Words: 16
150%
07:08 PM
19-May-25

sizeof("kishore" + 1);



sizeof(sizeof("Kishore"))



Bitwise operators

They work on bits[0, 1].

It is one of the low level feature of c.

They work with int data only.

C is a 16 bit compiler. Hence bitwise operators are taken from 2^0 to 2^{15} ➔ 2^{16}

When the starting bit is 1 it is a negative no. when starting bit 0, it is a +Ve no. hence the starting bit is also called it is a **sign** bit.

Eg:

00111110 ➔ +Ve

10111000 ➔ -Ve

We have to count only the 1's positions, not the 0's. any no * 0 gives 0.

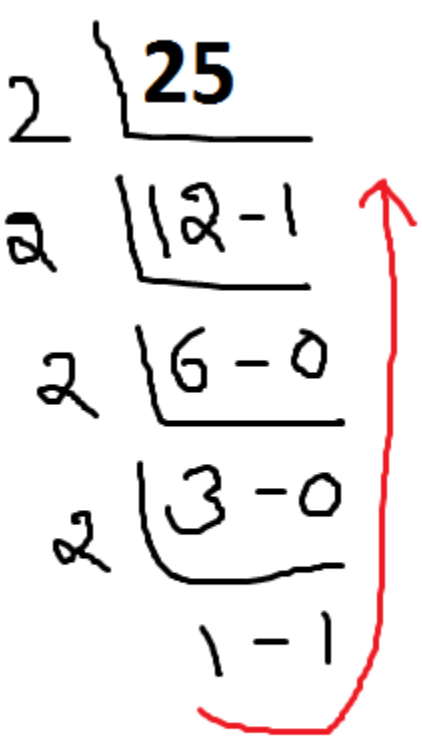
In c we are having 6 bitwise operators.

1. & - bitwise and
2. | - bitwise or
3. ^ - Xor – Exclusive or
4. ~ - compliment
5. << - left shift
6. >> - right shift

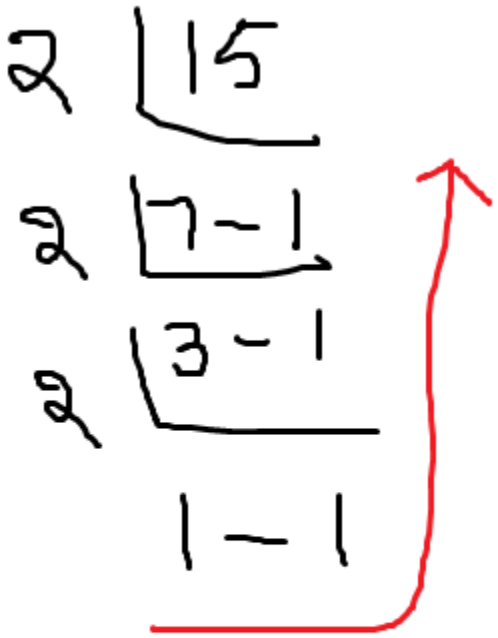
Eg:

& - when both bits are 1, result is 1.
Otherwise 0

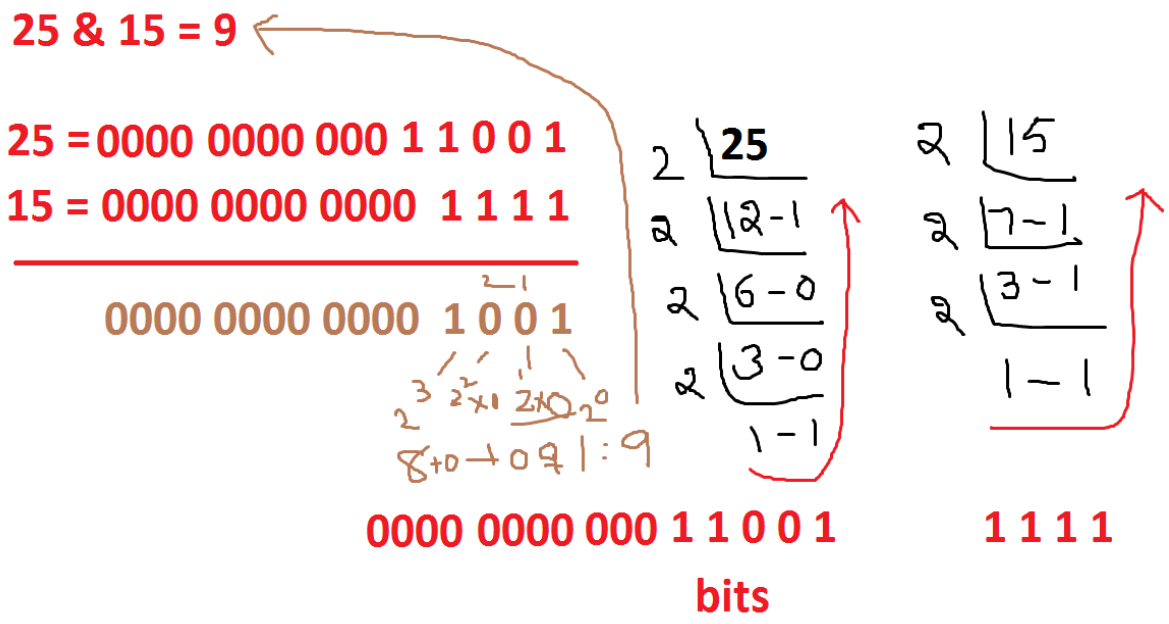
25 & 15 = 9



11001



1111


25 & 15 = 9
25 = 0000 0000 000 1 1 0 0 1
15 = 0000 0000 0000 1 1 1 1
0000 0000 0000 1 0 0 1
2 | 25
2 | 12 - 1
2 | 6 - 0
2 | 3 - 0
1 - 1
2 | 15
2 | 7 - 1
2 | 3 - 1
1 - 1
0000 0000 000 1 1 0 0 1
1 1 1 1
bits